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SUMMARY

Computer Science (AI/ML) undergraduate specializing in game development and AI-driven applications. Builds performance-optimized gameplay systems in Unreal Engine 5 / C++ and ships AI-powered web apps end-to-end, from architecture through deployment, under hackathon-speed timelines. Winner of the IEEE Vibe Code Hackathon (1st of 30+ teams).

TECHNICAL SKILLS

Game Development: Unreal Engine 5.6+, C++, Blueprint Scripting, AI Behaviour Trees, Enemy AI State Machines, NavMesh, Root Motion, Inverse Kinematics (IK), Chaos Physics

Web & Data: Supabase, HTML5/CSS3

AI / ML Integration: Gemini API, Tavily API, Prompt Engineering, Agentic Workflow Design

Tools & Performance: Git/GitHub, Visual Studio, Antigravity IDE, DirectX 11 Optimization, VRAM Management

Core CS: C++ (primary), OOP, Computer Graphics, Linear Algebra

KEY PROJECTS

Action-RPG Technical Demo

Unreal Engine 5 & C++

- Maintained a consistent 60 FPS during complex combat montages by engineering a frame-accurate movement system using Root Motion and Inverse Kinematics (IK).
- Eliminated VRAM-related crashes on mid-range hardware (RTX 3050, 6GB) by switching the default RHI to DirectX 11 and optimizing texture streaming.
- Architected modular combat logic in a hybrid C++/Blueprint system with custom animation montages and precise, AnimNotify-gated weapon collision detection.
- Built enemy AI as a state machine (Patrolling, Chasing, Attacking, Dead) with NavMesh patrol and perception-driven chase logic; added Chaos Physics breakable objects via a hit interface.

WebScout: Autonomous AI Web Agent for Product Research

Supabase, Gemini API, Tavily API

- Designed and shipped a full-stack autonomous agent solo within a fixed hackathon window, from system architecture through deployment.
- Integrated Gemini and Tavily APIs so the agent independently researches, compares, and summarizes products in real time from live web data.
- Built a Supabase-backed data layer and a structured 4-file reference workflow (architecture, design, tech stack, procedure) to stay fast under time pressure.

Isekai Dock: Windows 11 Taskbar Theme Mod

Windhawk, Taskbar Styler

- Designed 45+ custom taskbar themes (anime, fantasy, cyberpunk) with a curved centered dock and rounded tray; planning an open-source release for community themes.

Aizen: Artisan Fabric Marketplace

Developer

- Built a direct-to-consumer MVP digitizing the handicraft supply chain, removing middleman costs for local artisans.
- Developed a rule-based fabric estimation engine that calculates material needs and cost, reducing pricing errors.

Code Nest: Adaptive AI Programming Lab

System Architect

- Authored the system architecture for an AI-powered programming lab that personalizes CS education through adaptive, real-time code feedback modules.

ACHIEVEMENTS & HONORS

Winner, Vibe Code Hackathon (IEEE)

Feb 2026

- Won 1st place among 30+ teams with "Quick-Byte," a no-login digital kiosk system for high-traffic canteen ordering.
- Reduced checkout latency by architecting a real-time ordering workflow and a live kitchen dashboard synchronizing orders across preparation stations.

1st Place, IEEE Debate Competition

- Placed first as part of a 4-member team under competitive time constraints.

CERTIFICATIONS & COURSES

- Unreal Engine 5 C++: The Ultimate Game Developer Course** (Stephen Ulibarri, Udemy): completed Sept 2026
- C++ Programming** (Abdul Bari, Udemy): completed
- IBM SkillsBuild:** Artificial Intelligence, Generative AI

EDUCATION

Bachelor of Engineering, Computer Science (AI & Machine Learning)

Expected 2028

Saboo Siddik College of Engineering, Mumbai University | Social Media Joint Head, IEEE Student Chapter